

Practical - 3

Government College of Engineering, Jalgaon (An Autonomous Institute of Govt. of Maharashtra) Department of Computer Engineering	
Name:	
PRN:	Batch:
Class:	Semester:
Course:	Subject Teacher:
Date of Performance:	Date of Completion:

Aim: To study Message Passing Interface (MPI).

Title: To study Message Passing Interface (MPI) for Distributed Computing System.

Theory

Message Passing Interface (MPI)

Objective:

Message Passing Interface (MPI) is a standardized and portable message passing system designed for parallel and distributed computing. It provides a set of routines that can be efficiently implemented by hardware vendors.

MPI allows multiple computers (or processors) to communicate and exchange data while working on a common computational problem.

Introduction :

In parallel computing, multiple processors (called nodes) work simultaneously on different parts of a problem. These nodes must:

- Synchronize their actions
- Exchange data
- Maintain control over execution

MPI provides a standard set of functions to perform these tasks efficiently.

Features of MPI

1. General Features

- Communicators combine context and group for secure communication
- Thread safety support

2. Point-to-Point Communication

- Supports structured buffers and derived datatypes
- Works in heterogeneous environments

Modes:

- Blocking (Normal)
- Non-blocking
- Synchronous
- Ready mode (for faster protocols)
- Buffered mode

3. Collective Communication

- Built-in and user-defined collective operations
- Supports large data movement routines
- Allows subgroup creation (by topology or manually)

4. Not Included in MPI

- Process management
- Remote memory transfer
- Active messages
- Threads
- Virtual shared memory

5. Compatibility

MPI does not directly implement the above concepts but remains compatible with them (e.g., thread safety and intercommunicators).

Working of MPI in Distributed Computing

1. Distributed Memory

- Each processor has its own local memory
- Memory is not shared directly
- Data transfer happens over a network
- Different from shared memory systems

2. Message Passing

- Data is transferred as packets between processors
- Messages include control and routing information
- Can consist of one or more packets

3. Process

- A process is a running program
- Multiple processes can run on a processor
- In MPI, processes communicate via messages

4. Message Passing Library

- Collection of routines used for communication
- Includes send, receive, and synchronization functions

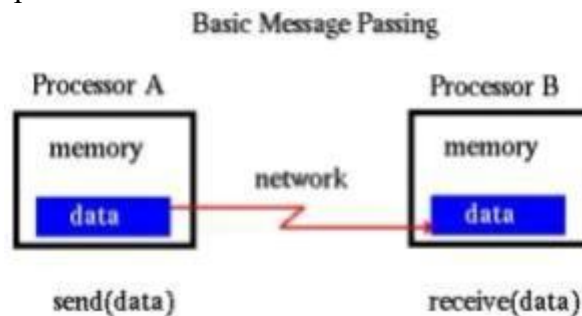
Send and Receive Operations

- **Send:** Transfers data from one process
- **Receive:** Accepts data in another process

Requirements:

- Data location
- Data size
- Data type
- Destination

Both send and receive operations must match.



Communication Types

1. Synchronous Communication

- Sender waits until receiver acknowledges

2. Asynchronous Communication

- Sender continues without waiting

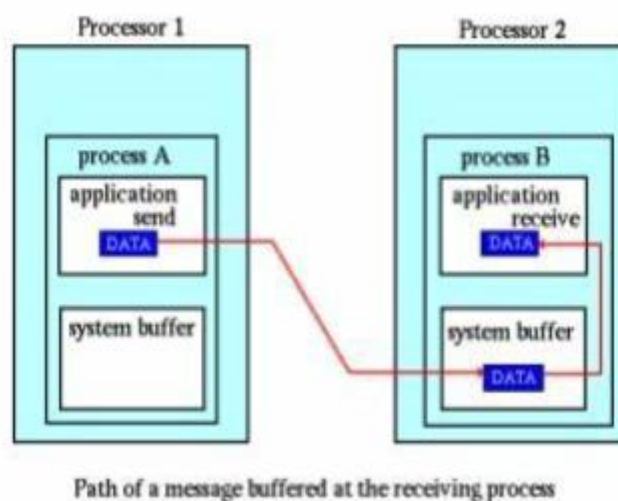
Buffers

1. Application Buffer

- Memory area where program data resides

2. System Buffer

- Temporary storage used during communication
- Enables asynchronous communication



Socket-Based Communication Concepts

1. Socket acts as an endpoint for communication
2. Uses a two-level naming scheme
3. Supports location transparency
4. Provides flexibility using typed communication

IPC Primitives

1. Socket Creation

`s = socket(domain, type, protocol)`

2. Bind

`bind(s, addr, addrlen)`

3. Connect

`connect(s, server_addr, server_addrlen)`

4. Listen

`listen(s, backlog)`

5. Accept

`snew = accept(s, client_addr, client_addrlen)`

6. Sending and Receiving Data

`read(snew, buffer, amount)`

`write(s, "message", msg_length)`

`recvfrom(s, buffer, sender_address)`

`sendto(s, "message", receiver_address)`

Conclusion

In this practical, we studied the Message Passing Interface (MPI) and understood its role in enabling communication between processes in distributed computing systems.

Sign of teacher
Mrs. Archana Chitte.